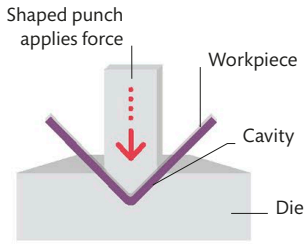
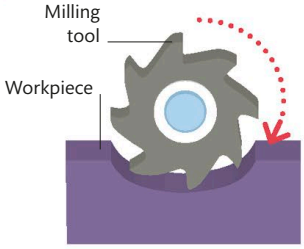
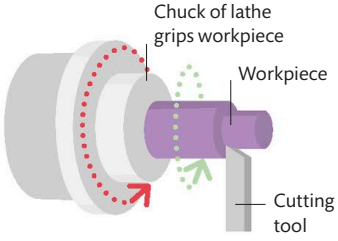
COLD METHODS



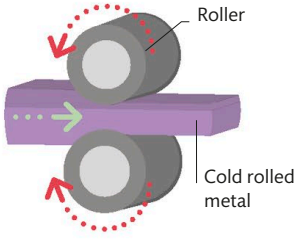
Bending
Many products are made by a process that is also known as cold forging, in which pressure is applied to force a metal workpiece into a cavity to achieve the desired shape.



Milling
A milling machine shapes the workpiece by wearing away the excess parts with a milling bit. During the process, the machine sprays both bit and metal with a coolant.



Turning
The workpiece is shaped by a fixed cutting tool while being rotated by a lathe. Turning can produce only objects that are symmetrical around the axis of rotation.



Rolling
The metal is shaped by rollers. Sheets, strips, bars, and rods are cold rolled to obtain products that have smooth surfaces and accurate dimensions.

5,700°F (3,150°C)—THE TEMPERATURE IN THE **FLAMES** OF SOME **OXYACETYLENE TORCHES**

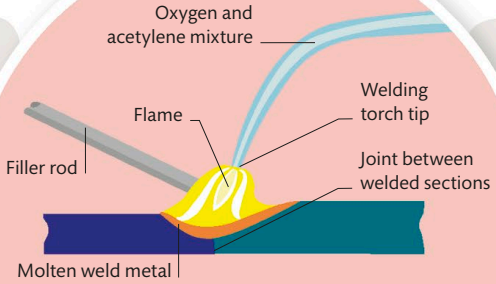
TREATING METALS

Metals can be treated in different ways to adapt their properties. Some common treatments aim to make the metal less brittle, while others prevent rust and corrosion.

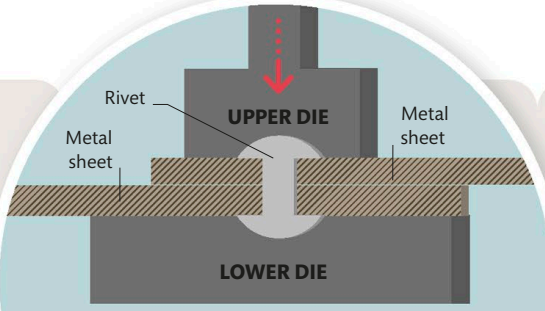
Tempering
The metal is heated to a specific temperature and allowed to gradually cool. The process removes hardness and increases the toughness.

Anodizing
Metal is submerged into an electrolytic solution through which a current is passed. This forms a metal oxide film that increases corrosion resistance.

Galvanizing
The metal is submerged into a bath of molten zinc, resulting in a protective zinc coating that prevents rusting.



Welding
In welding, the two pieces of metal are heated by a localized heat source and melted along the joint. A filler with a similar melting point can be used to strengthen the joint.



Riveting
A rivet is a shaft of metal. It is placed into a premade hole, and the heads of the rivet are then mechanically compressed into domes by two dies. For large structures, bolts are usually used instead of rivets.